Computer Organization

Lab6 Integer Arithmetic

Integer Arithmetic

Topics

- Arithmetic
  - Adder
  - Subtraction
  - Multiplication
  - Division
- Practice

Adder with overflow detector (1)

- The rule about overflow if result out of range on adder:
  - ✓ no overflow, if adding +ve and -ve operands
  - ✓ overflow, if
    - adding two +ve operands, get -ve operand 001 + 011 = 110
    - adding two -ve operands, get +ve operand 101 + 101 = (1) 010

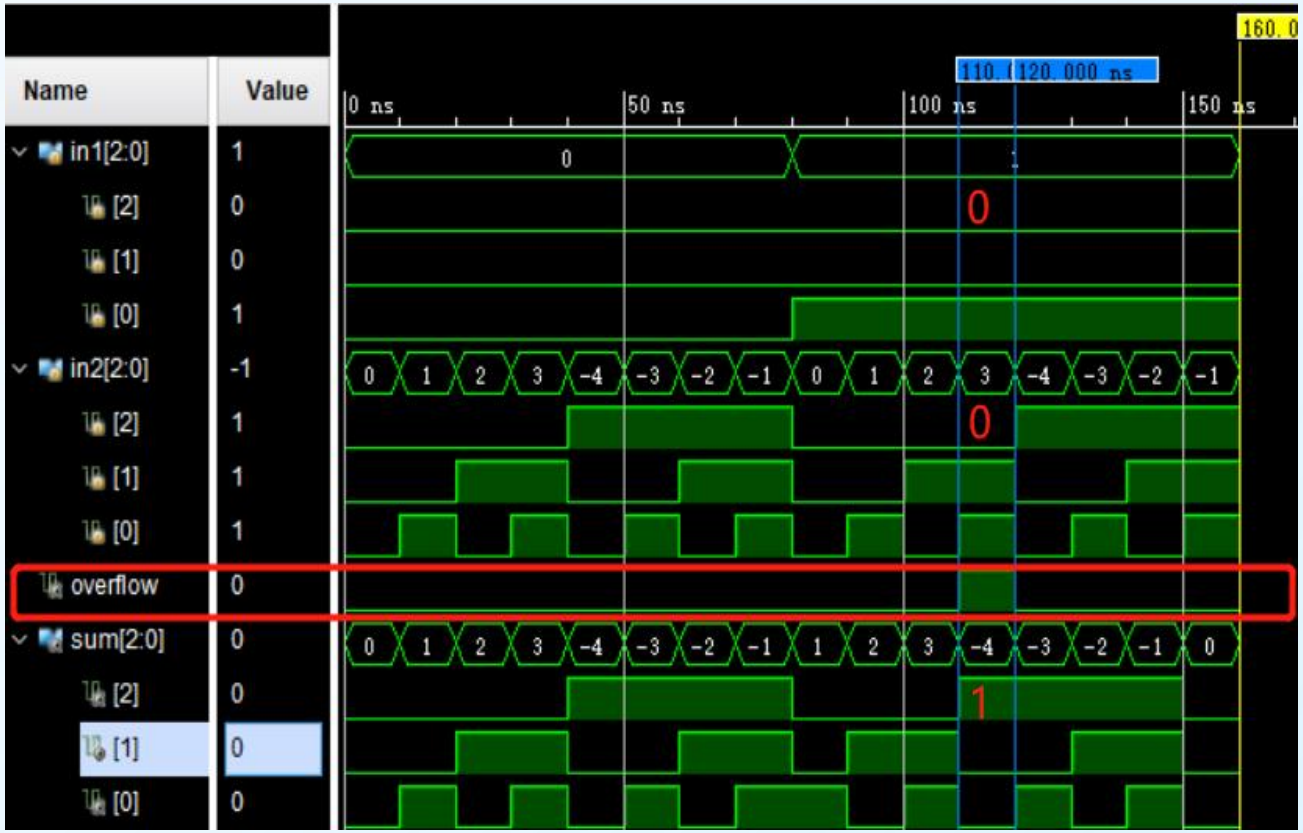
```
module adder (in1, in2, sum, overflow);                    //in verilog
input [2:0]in1, in2;
output [2:0] sum;
output overflow;

assign sum = in1 + in2;
assign overflow =
 ( in1[2] &   in2[2] & ~sum[2] )    |    /* two +ve operands, get -ve operand*/
 (~in1[2] & ~in2[2] &   sum[2] );       /* two -ve operands, get +ve operand*/

endmodule
```

Adder with overflow detector (2)

- The figure is the waveform of the circuit “adder”
  - in1 is the addend and in2 is the addend.
  - while the value of overflow is 1’b1, it means there is a overflow, otherwise means not.



- Observe the waveform from 110ns to 120ns
- the in1 is 3’b001 and in2 is 3’b011, the sum is 3’b100
- The signed bit of in1 and in2 is 0 (means they are both +ve), the singed bit of sum is 1(means it is -ve)
- In this situation, an overflow is detected

Subtraction (1)

- Implement the subtraction with adder: add negation of second operand
- How to get the negation of a number? Which of the following option(s) is(are) right?
  - ✓ Option1:
    - step1: Invert the sign bit. e.g. vin2p1 = ~in2[2]
    - step2: add 1 after inverting the value bits. e.g. ~in2[1:0] + 1
  - ✓ Option2:
    - Inverting the bits and adding on1. e.g. ~in2+1

```
module subO1(in1,in2,result);   //verilog
input [2:0] in1;
input [2:0] in2;
wire vin2p1;
wire [1:0] vin2p2;
output [2:0] result;
assign vin2p1 = ~in2[2];
assign vin2p2 = ~in2[1:0] + 1;
assign result = in1 + {vin2p1,vin2p2};
endmodule
```

```
module subO2(in1,in2,result);   //verilog
input [2:0] in1;
input [2:0] in2;
output [2:0] result;
wire [2:0] vin2;
assign vin2= ~in2 + 1;
assign result = in1 + vin2;
endmodule
```

Subtraction (2)

- Verify the function of the circuit ‘subO1’, ‘subO2’, Which implement(s) of sub is(are) correct?

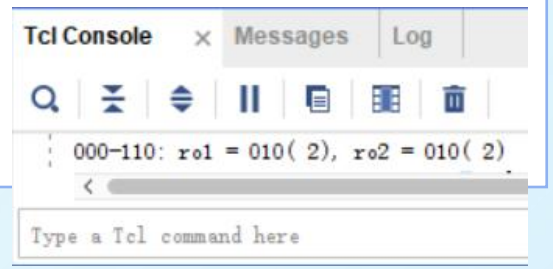
```
module subO1(in1,in2,result);    //verilog
input [2:0] in1;
input [2:0] in2;
wire vin2p1;
wire [1:0] vin2p2;
output [2:0] result;
assign vin2p1 = ~in2[2];
assign vin2p2 = ~in2[1:0] + 1;
assign result = in1 + {vin2p1,vin2p2};
endmodule
```

```
module subO2(in1,in2,result);    //verilog
input [2:0] in1;
input [2:0] in2;
output [2:0] result;
wire [2:0] vin2;
assign vin2= ~in2 + 1;
assign result = in1 + vin2;
endmodule
```

```
module subTb( );          //verilog
reg [2:0] in1,in2;
wire [2:0] rO1, rO2;
subO1  usubO1(in1,in2,rO1);
subO2  usubO2(in1,in2,rO2);

initial begin
   {in1,in2} = 6'b0;
   $monitor( "%3b-%3b: ro1 = %3b(%d), ro2 =
%3b(%d)",
in1,in2,rO1,$signed(rO1) ,rO2,$signed(rO2) );
   repeat(63)   #10 {in1,in2} = {in1,in2} + 1;
   #10 $finish();
end

endmodule
```



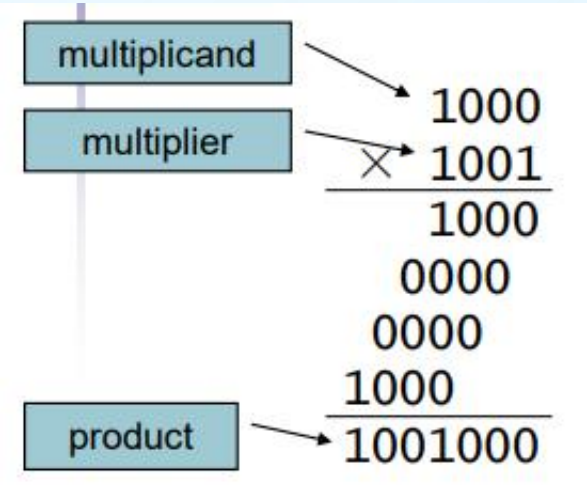
TIPs:
$monitor is a system service in **verilog**, which is valid only in **simulation**. It monitor the datas: whenever any of them changes, it prints the datas in the specified format.
In Vivado, the print information is showed in tcl window.

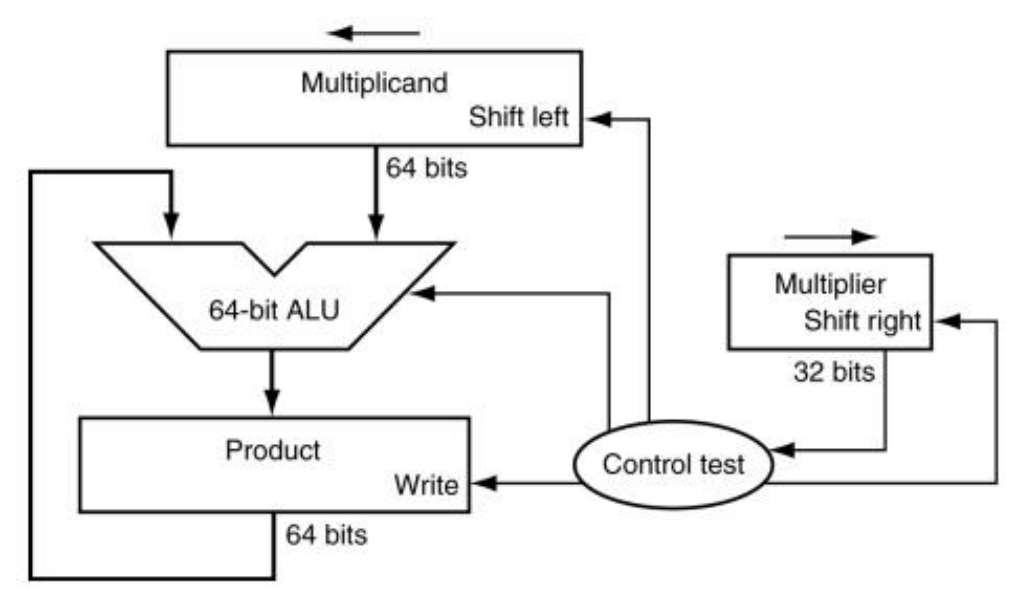
%3b: means print the data in **binary**, the bitwidth is **3**
%d: means print the data in **decimal**

$signed is a system service in verilog, which change the data tobe signed value.

Multiplication (1)

- Here is a digital circuit which implement the long-multiplication approach:
  - ✓ Shift registers for multiplicand and multiplier
    - store and shift
  - ✓ Adder with two inputs and a control signal
    - add or not
  - ✓ A register to store the product
    - when to get the data from the product register ?





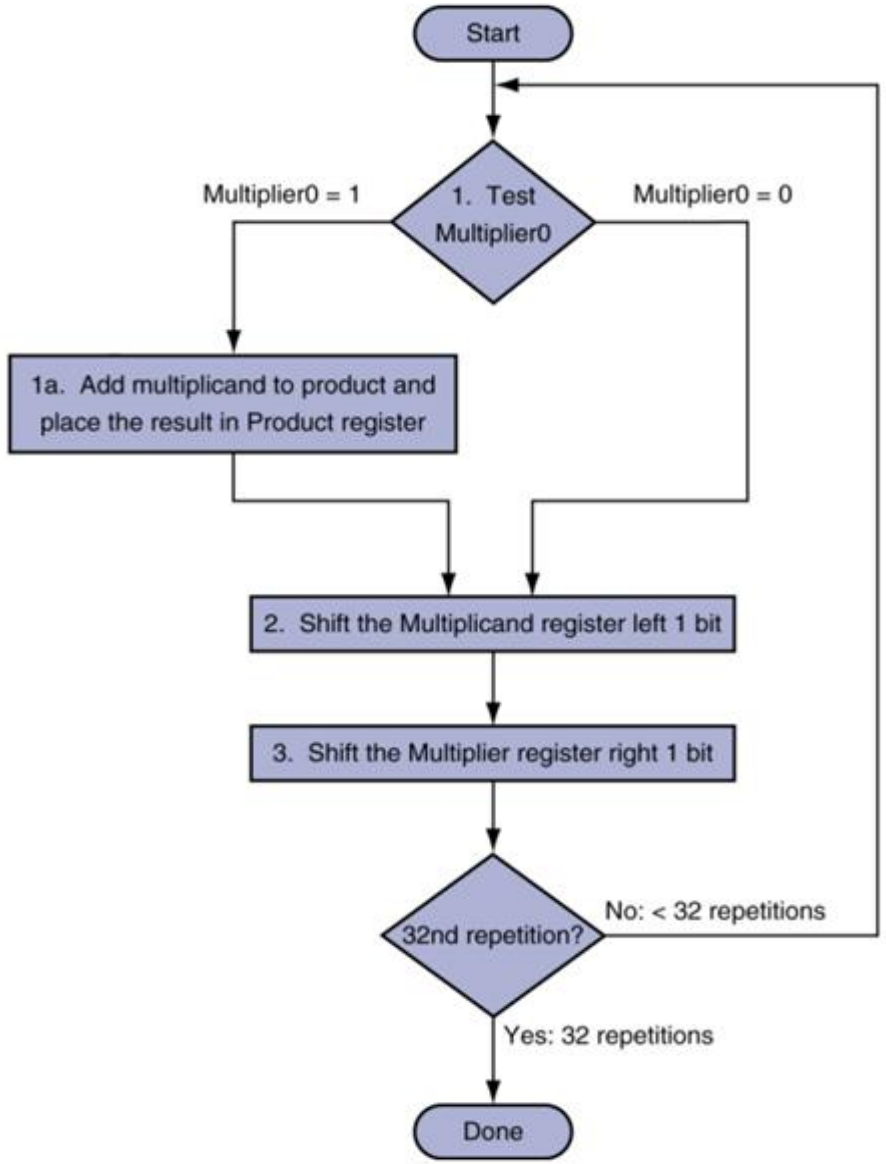
Multiplication (2)

- Can this piece of codes get the correct product result?
- If the length of multiplier is less than 4, could the assembly code work more effectively ?

```
# Piece 6-1
.data
        m1:.byte 8        # multiplicand
        m2:.byte 9        # multiplier
.text
        lb t0, m1
        lb t1, m2
        add t2, zero, zero
loop:
        li s1, 1
        and s2, s1, t1    #to determine the lowest bit of t1
        beq s2, zero, jumpAdd
        add t2, t0, t2
jumpAdd:
        slli t0, t0, 1
        srli t1, t1, 1
        addi a0, a0,1
        li a1, 4          # 4 is the length of 9 in binary
        blt a0, a1, loop

        mv a0, t2
        li a7, 1
        ecall

        li a7, 35
        ecall
```



Multiplication instructions in RISC-V

- RV32M multiply extension
- mul, mulh, mulhsu, mulhu

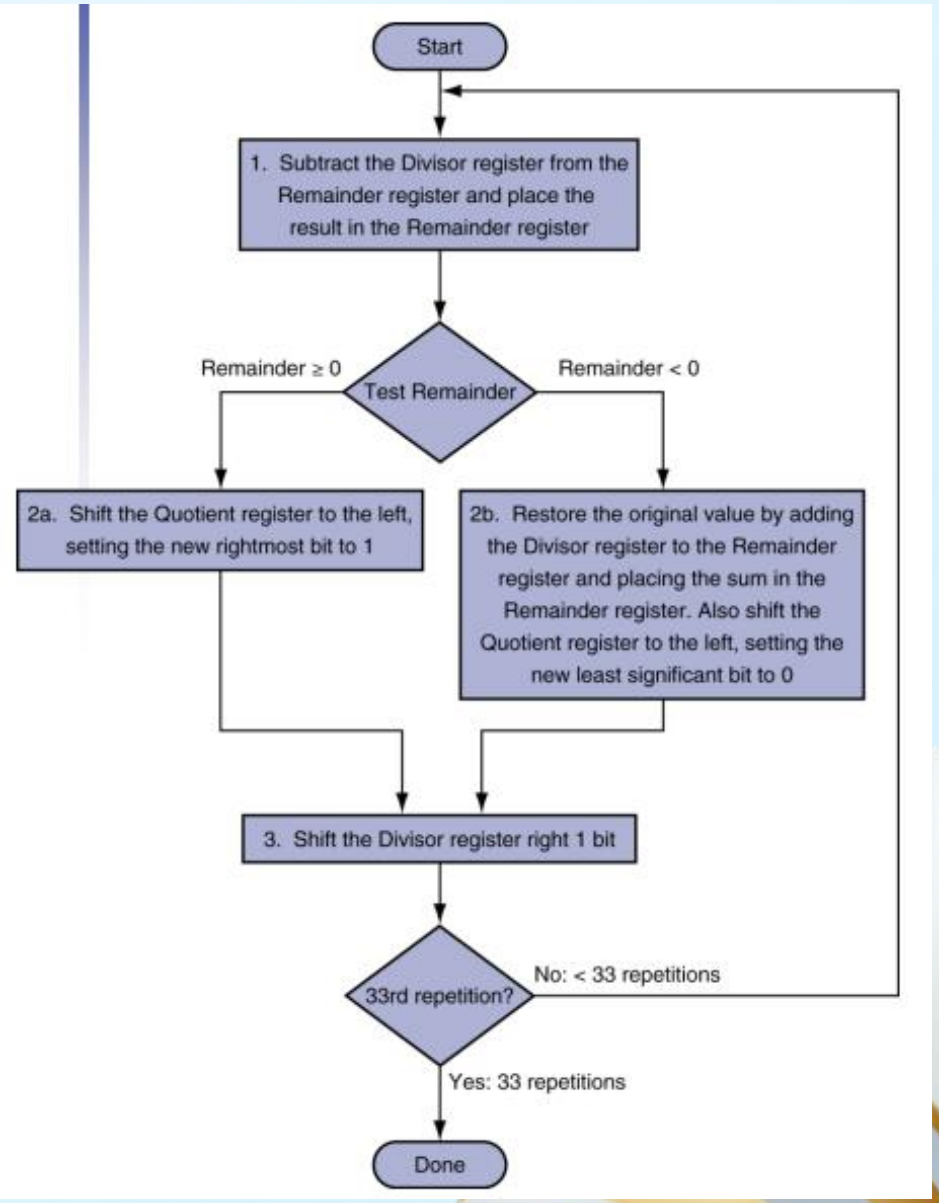
Inst	Name	FMT	Opcode	funct3	funct7	Description (C)
mul	MUL	R	0110011	0x0	0x01	rd = (rs1 * rs2)[31:0]
mulh	MUL High	R	0110011	0x1	0x01	rd = (rs1 * rs2)[63:32]
mulhsu	MUL High (S) (U)	R	0110011	0x2	0x01	rd = (rs1 * rs2)[63:32]
mulhu	MUL High (U)	R	0110011	0x3	0x01	rd = (rs1 * rs2)[63:32]

- If both high and low 32-bit multiplication product are required, the general code order is:

```
mulh rdh, rs1, rs2   #or mulhsu rdh, rs1, rs2  or mulhu rdh, rs1, rs2
mul rdl, rs1, rs2
```

Division (1)

- Check for 0 divisor by software.
- Long division approach
  - ✓ If divisor ≤ dividend bits: 1 bit in quotient, subtract
  - ✓ Otherwise: 0 bit in quotient, bring down next dividend bit
- Restoring division
  - ✓ Do the subtract, and if remainder goes < 0, add divisor back
- Signed division
  - ✓ Divide using absolute values
  - ✓ Adjust sign of quotient and remainder as required



Division (2) long division approach

- Step0: prepare for the long division approach

```
# Piece 6-3
# Piece 6-3-1
.include "macro_print_str.asm"
.data
        dividend: .word 7
        divisor:  .word 2
        x: .word 0x8000
        looptimes: .byte 5
.text
        la t5, dividend
        lw t1, (t5)    # t1 : diviend
        la t5, divisor
        lw t2, (t5)    # t2 : divisor
        slli t2, t2, 4
        la t5, dividend
        lw t3, (t5)  # t3 store the remainder
        add t4, zero, zero     # t4 quotient

        la t5, x
        lw a1, (t5)            #a1 used to get the highest bit of remainder
        add t0, zero, zero   # t0: loop cnt
        la t5, looptimes
        lb s1, (t5)            #s1: looptimes
```

- Divide 7_{dec} ($0000\ 0111_{bin}$) by 2_{dec} (0010_{bin})

Iter	Step	Quot	Divisor	Remainder
0	Initial values	0000	0010 0000	0000 0111
1	Rem = Rem – Div	0000	0010 0000	1110 0111
	Rem < 0 ➔ +Div, shift 0 into Q	0000	0010 0000	0000 0111
	Shift Div right	0000	0001 0000	0000 0111
2	Same steps as 1	0000	0001 0000	1111 0111
		0000	0001 0000	0000 0111
		0000	0000 1000	0000 0111
3	Same steps as 1	0000	0000 0100	0000 0111
4	Rem = Rem – Div	0000	0000 0100	0000 0011
	Rem >= 0 ➔ shift 1 into Q	0001	0000 0100	0000 0011
	Shift Div right	0001	0000 0010	0000 0011
5	Same steps as 4	0011	0000 0001	0000 0001

Division (3) long division approach

- Step1-5: Do the long division approach

```
# Piece 6-3-2
loopb:
        # t1: dividend,   t2: divisor,    t3: remainder,      t4: quotient
        # a1: 0x8000,  s1: 5

        sub t3, t3, t2              #dividend  - divisor
        and s0, t3, a1              # get the highest bit of remainder to check if rem<0
        slli t4, t4, 1          # shift left quot with 1bit
        beq s0, zero, SdrUq     # if  rem>=0, shift Div right
        add t3, t3, t2              # if rem<0, rem=rem+div
        srli t2, t2, 1
        addi t4, t4, 0
        j loope

SdrUq:
        srli t2, t2, 1
        addi t4, t4, 1

loope:
        addi t0, t0, 1
        bne t0, s1, loopb
```

```
# Piece 6-3-3
        li a7, 1
        mv a0, t4  #print quotient
        ecall
        print_string("\n")
        li a7, 1
        mv a0, t3 #print remainder
        ecall

        li a7, 10
        ecall
```

- Divide 7_{dec} ($0000\ 0111_{bin}$) by 2_{dec} (0010_{bin})

Iter	Step	Quot	Divisor	Remainder
0	Initial values	0000	0010 0000	0000 0111
1	Rem = Rem – Div	0000	0010 0000	1110 0111
	Rem < 0 ➔ +Div, shift 0 into Q	0000	0010 0000	0000 0111
	Shift Div right	0000	0001 0000	0000 0111
2	Same steps as 1	0000	0001 0000	1111 0111
		0000	0001 0000	0000 0111
		0000	0000 1000	0000 0111
3	Same steps as 1	0000	0000 0100	0000 0111
4	Rem = Rem – Div	0000	0000 0100	0000 0011
	Rem >= 0 ➔ shift 1 into Q	0001	0000 0100	0000 0011
	Shift Div right	0001	0000 0010	0000 0011
5	Same steps as 4	0011	0000 0001	0000 0001

Division instructions in RISC-V

- mul, mulh, mulhsu, mulhu

Inst	Name	FMT	Opcode	funct3	funct7	Description (C)
div	DIV	R	0110011	0x4	0x01	rd = rs1 / rs2
divu	DIV (U)	R	0110011	0x5	0x01	rd = rs1 / rs2
rem	Remainder	R	0110011	0x6	0x01	rd = rs1 % rs2
remu	Remainder (U)	R	0110011	0x7	0x01	rd = rs1 % rs2

- If both quotient and remainder are required, the general code order is:

```
div rdq, rs1, rs2   # or divu rdq, rs1, rs2
rem rdr, rs1, rs2   # or remu rdr, rs1, rs2
```

Practice 1-1: Adder with overflow detector

- Please complete the test bench to finding all of the legal combinations of the two inputs, verify the function of the circuit “adder”.
- A test bench on the bottom is for the reference.

```
module adderTb( );     //verilog
reg [2:0] in1,in2;
wire overflow;
wire [2:0] sum;

adder  ua(in1,in2,sum,overflow);

initial begin
   {in1,in2} = 6'b0;
   repeat(15) #10 {in1,in2} = {in1,in2} + 1;
   #10 $finish;
end

endmodule
```

Practice 1-2: Subtraction with overflow detector

- Please complete the circuit to detect the overflow of the subtraction.
- Build a test bench to verify the function of the circuit.
- The description about the overflow of the subtraction is described as bellow:

- Overflow if result out of range
  - No overflow, if subtracting two +ve or two –ve operands
  - Overflow, if:
    - Subtracting +ve from –ve operand, and the result sign is 0 (+ve)
    - Subtracting –ve from +ve operand, and the result sign is 1 (-ve)

```
//verilog
module subtraction(in1,in2,result,overflow);

input [2:0]in1,in2;
output [2:0] result;
output overflow;

assign result = in1 - in2;
assign overflow = ______________________;

endmodule
```

Practice 2-1: Multiplication

- The assembly code on the right hand is just for the multiplier whose width is **not larger than 4**, and only for the **unsigned multiplication**, modify the code to achieve the following function:
  - ✓ 1) The width of multiplicand and multiplier is 16.
  - ✓ 2) The highest bit is take as the sign bit, to implement the signed multiplication.
  - ✓ Note: Don't use the mul instruction.

```
# Piece 6-1
.data
        m1:.byte 8        # multiplicand
        m2:.byte 9        # multiplier
.text
        lb t0, m1
        lb t1, m2
        add t2, zero, zero
loop:
        li s1, 1
        and s2, s1, t1   #to determine the lowest bit of t1
        beq s2, zero, jumpAdd
        add t2, t0, t2
jumpAdd:
        slli t0, t0, 1
        srli t1, t1, 1
        addi a0, a0,1
        li a1, 4            # 4 is the length of 9 in binary
        blt a0, a1, loop

        mv a0, t2
        li a7, 1
        ecall

        li a7, 35
        ecall
```

Practice 2-2: Division

- The piece of assembly code of piece 6-3 on Page 13 and 14 is just for the **8 bit unsigned division**. Take piece 6-3 as reference, do the following tasks:
- 1) To implement a 32 bit division with detecting exception while the divisor is 0.
- 2) The highest bit is taken as the sign bit, to implement the signed division.
- For signed division:
  - ✓ Step1: Divide using absolute values.
  - ✓ Step2: Adjust sign of quotient and remainder as required.
  - ✓ The quotient is “+”, if the signs of divisor and dividend agrees, otherwise, quotient is “–”.
  - ✓ The sign of the remainder matches that of the dividend.
  - ✓ (+7) ÷ (-2) = (-3) ··· (+1) (-7) ÷ (-2) = (+3) ··· (-1)
- Note: Don’t use the div instruction.